

ATLAS PDD — ATANDA Command Centre

Drafted with the F6 ATLAS PDD Drafter methodology (the platform's own 4-Part ATLAS Project Definition Document format), `mode=FROM_SPC` against the live codebase as the source-of-truth "SPC". This document captures the present system state **plus the current ARK.ONECRAFT JST-import changes**.

- **Schema:** `atlas-pdd-v1` · **Source:** present monorepo HEAD
- **Parts:** Cheat Sheet · Exec Summary · Worksheet (ROYGBIV) · Implementation
- Appendix A is the **ATLAS-J crystallised** typed view (phase rainbow, CLASS A/B/C).

Part 1 — Cheat Sheet

1-page distilled pitch. Audience: founders.

What we're building. The ATANDA Command Centre — an authenticated subscription portal in front of the **FORGE.BONSAI HARNESS**. The HARNESS is a PDD-blueprint application: an ordered sequence of atomic prompts (the instruction layer for an LLM), **not** an SPC. Per session a creator runs one coherent pipeline — **F1 → F2 → F3 → F4 → F5 → F6 (+VIBE DJ) → F7**, with **F8 Code DJ** scaffolding a runnable codebase from the certified spec. Side-step engines (**ATLAS J**, **PFP drift detector**, **DE-SPC synthesiser**) and an onboarding spine (**Ascension Protocol**, anchored to The Atomic Prompt) wrap the core.

For whom. Solo creators, practitioners, and architects who hold a raw idea and want a deterministic path from prompt → SPC → PDD → SPARTAN-certified MVP-PDD → runnable code. Teams buy per-seat subscriptions that confer elevated tiers.

Why now. The JST-displacement thesis (Jobs · Skills · Talent) is the wedge: the platform meets a creator at their "number" and climbs them to a shipped, verifiable system. The new **ARK.ONECRAFT integration** turns that number from a self-reported estimate into an authoritative, externally-sourced score.

Top 3 decisions. 1. **Contract-first.** `lib/api-spec/openapi.yaml` is the single source of truth; Zod + React Query are generated (operationId-shaped). 2. **App-layer authorization, no RLS.** Every query is scoped by `req.localUser.id` / `req.effectiveTier`; Clerk (Replit-managed) is the identity bridge. 3. **Honest-failure integrations.** External integrations (Sphinx, ARK.ONECRAFT) are env-gated and return a typed `503 * _NOT_CONFIGURED` rather than fabricating data — the interim in-app path stays usable until creds are set.

Success criteria (measurable). - A signed-in user completes F1→F7 and obtains a public verification URL. - Runaway-spend is bounded: every engine route enforces a monthly USD cost cap. - Cross-provider engine output is reproducible: 30-test fixture suite runs offline. - ARK.ONECRAFT score, once connected, supersedes self-assessment with **zero code change** (newest JST row wins).

Out of scope (this PDD). Re-platforming away from Express/Drizzle/Clerk/Stripe; the receiver-side ARK.ONECRAFT service implementation (we own only the client).

Part 2 — Executive Summary

Business-outcome story. Audience: investors / stakeholders.

Market context. AI-assisted building is crowded with one-shot prompt tools, but they stop at "generates something". The Command Centre productises rigour: a graded pipeline that certifies an idea (JCSE scoring, SPARTAN certification, PFP drift gating) before code is scaffolded. The output is auditable, not just plausible.

Competitive frame. Where "VIBE" tools (Cursor, Lovable, Bolt, v0) optimise the build step, the HARNESS optimises everything upstream of it — and then **recommends the right VIBE** (F6-VDJ) for the certified spec. The platform is therefore complementary to, not competing with, the IDE/agent layer.

Stakeholder map. - **Creators / Practitioners / Architects** — primary subscribers (tiered). - **Teams / Institutions** — per-seat subscriptions (`team` → INSTITUTION, `team_lite` → ARCHITECT) with org tenancy, audit logs, weekly digests. -

ARK.ONECRAFT — sister platform; authoritative source of the JST score and the Sphinx Marketplace publish target. - **Ops / Admin** — cost-cap overrides, scheduled cron ticks, deployment.

Milestones (month markers). - **M0 — Core pipeline shipped.** F1-F8 + ATLAS J + PFP + DE-SPC live; portal shell, pricing, Stripe tiers, per-seat orgs, activity log, cost dashboard all in place. - **M1 — Onboarding spine (Ascension Protocol).** JST primitive, reader track, 13-rung book-anchored ladder composing real signals. (merged) - **M2 — ARK.ONECRAFT JST import (current change).** `source`-tagged JST scores; env-gated import endpoint; front-end upload CTA framing self-assessment as interim. - **M3 — Federation.** Broaden Sphinx + ARK.ONECRAFT into a two-way marketplace and identity federation (VIOLET phase, deferred).

Investment ask (if relevant). The remaining spend is integration go-live (ARK.ONECRAFT receiver, Sphinx receiver) and the enterprise hardening already scaffolded (SSO/RBAC/audit are in place; multi-region is deferred). No core rebuild is required.

Part 3 — Worksheet

Phase-by-phase build plan using the canonical ATLAS phase rainbow. Each phase honours its fixed thematic meaning. MVP ships RED→BLUE; INDIGO is largely shipped; VIOLET/WHITE deferred.

■ RED — Integration Foundation · shipped

Duration: M0. DB, auth, API proxies, secrets, base runtime. - PostgreSQL + Drizzle ORM (one schema file per table, barrel re-export). - Clerk (Replit-managed) proxy mounted **before** body parsers; auth bridge JITs a local user + subscriber row (`req.localUser` , `req.effectiveTier`). - Stripe wired (raw webhook route mounted before JSON parser; idempotency via `stripe_webhook_events` PK). - Anthropic via Replit AI Integrations proxy (Claude Sonnet 4). - **Exit criterion:** `curl localhost:80/api/healthz` green; authed request resolves a subscriber; Stripe webhook replay returns `{ok:true, replay:true}`.

■ ORANGE — Portal Shell · shipped

Duration: M0. Dashboards, navigation, pricing UI, session shell. - `command-centre` artifact serves `/` landing + `/pricing` ; tiered Stripe checkout (Practitioner/Architect monthly+yearly, per-seat team + team-lite). - Session shell + session-detail page driving each engine; TopNav with Costs, Quests, Ascension, Cartridge links. - **Exit criterion:** a user can subscribe, land in the portal, and open a session.

■ YELLOW — Feature Workspaces · shipped

Duration: M0-M2. The productised engines operators actually use. - **HARNESS engines F1-F8** with per-engine workspaces; F8 Code DJ (Architect-tier) emits a `CODEBASE_BUNDLE` (≤ 12 files + manifest). - **ATLAS J** (engineId 10) crystallises an `ATLAS_PDD` into typed `ATLAS_PDD_JSON` . - **PFP** (engineId 11) drift detector; F8 enforces a hard `DRIFT_GATE` (HTTP 409) unless `acknowledgeDrift:true` . - **DE-SPC** evolve synthesiser. - **Cartridge** (\$499.99/project) + **Ingestion** (per-project credit) intake paths; cartridge context is an authoritative instruction layer prepended to every prompt. - **Ascension Protocol** onboarding ladder + **JST primitive**. - **ARK.ONECRAFT JST import (current change):** "UPLOAD JST SCORE FROM ARK.ONECRAFT" CTA on `/ascension` ; `source`-tagged scores; newest row supersedes. - **Exit criterion:** F1→F7 completes end-to-end and yields a verification URL; ARK import returns the typed not-configured signal in the absence of creds.

■ GREEN — CMS & Marketing · shipped

Duration: M0. Content seed, SEO, landing, verify pages. - Public `/pricing` , exemplar library with "Fork to session" CTA, public SPARTAN verification URL per certified MVP-PDD. - Email dispatch (Resend) with `[email:dry-run]` console fallback when unset. - **Exit criterion:** verification URL renders publicly; pricing is live.

■ BLUE — QA & Security · shipped

Duration: M0-M2. E2E tests, audit, threat model, launch gate. - Cross-provider engine suite (30 tests) replays cached LLM fixtures (~15s, offline); nightly `test:live` GitHub Action refreshes fixtures + opens a drift issue. - **Activity / audit log** (own + org-scoped, CSV export, cross-org leak guard). - **Runaway-spend guard:** `requireCostBudget` after `rateLimit` on every engine route; `402 COST_CAP_EXCEEDED` ; live monthly SUM, fail-open on DB blip. - SSRF-hardened external fetches (badge verifier; ARK import uses an env-driven host). - **Exit criterion:** `pnpm run typecheck` + test suite green; cost cap enforced.

▣ INDIGO — Enterprise • largely shipped

Duration: M0–M1. SSO, RBAC, org tenancy, audit logs. - Organizations + members (`owner|admin|member` , last-owner-protected) + tokenised invites; per-seat subscriptions with `effectiveTier = max(personal, conferred)` . - Notifications: weekly digest (cron), billing-failure alerts, high-cost alerts — all fire-and-forget so a Resend outage never breaks a webhook ack. - **Exit criterion:** a team-seat holder transparently gets elevated F8 limits.

▣ VIOLET — Federation • deferred to v2

Multi-region, partner integrations, marketplace. - Sphinx Marketplace **publish** path is shipped (outbound); the two-way federated marketplace + ARK.ONECRAFT identity federation are deferred.

▣ WHITE — Operating System • deferred

Open API, plugin ecosystem, white-label. Not in current scope.

Part 4 — Implementation

Technical implementation document. Audience: engineers.

Stack

- **Monorepo:** pnpm workspaces, Node.js 24, TypeScript 5.9 (composite libs via `tsc --build` ; leaf artifacts `--noEmit`).
- **API:** Express 5 + Clerk (Replit-managed) + Stripe + pino.
- **DB:** PostgreSQL + Drizzle ORM. **No RLS** — authorization is app-layer via `req.localUser.id` / `req.effectiveTier` .
- **Validation:** Zod (`zod/v4`) + `drizzle-zod` . **API codegen:** Orval (operationId-shaped names — `CreateSessionBody` , not `SessionInput`).
- **Build:** esbuild (CJS). **LLM:** Claude Sonnet 4 via `@workspace/integrations-anthropic-ai` .
- **Frontend:** React + Vite, wouter, shadcn/ui, generated React Query hooks.

Data schemas (current — including the ARK.ONECRAFT change)

```
// lib/db/src/schema/jst-assessments.ts (current change in **bold** below)
export const JST_SOURCES = ["self_assessment", "ark_onecraft"] as const;
export type JstSource = (typeof JST_SOURCES)[number];

jst_assessments {
  id          uuid pk
  userId      text      // scoped owner
  jobsScore   integer 1..10
  skillsScore integer 1..10
  talentScore integer 1..10
  composite   numeric    // round(((sum-3)/27)*100, 2) → 0..100
  band        varchar    // SEEKER <40 · BUILDER <60 · OPERATOR <80 · ASCENDANT
  source       varchar(24) // NEW — default 'self_assessment'
  notes       text?
  createdAt   timestampz  // newest row drives band/rungs ⇒ ARK supersedes
}
```

Other notable tables: `harness_sessions` / `harness_artifacts` / `harness_engine_runs` (cost ledger, `user_created_idx` powering the live cost SUM), `organizations` / `organization_members` / `organization_invites` , `command_centre_subscribers` (`monthly_cost_cap_usd_override`), `integration_credentials` (AES-256-GCM), the cartridge + ingestion credit tables, badge tables, and `reader_onboarding` .

API surface (the ARK.ONECRAFT change)

```
POST /me/jst/import-from-ark      operationId: importJstFromArk   tag: onboarding
- requireAuth
- env-gated: 503 { code: "ARK_NOT_CONFIGURED" } when ARK_ONECRAFT_BASE_URL
  or ARK_ONECRAFT_API_KEY is unset (mirrors the SPHINX precedent – never
  fabricates a score)
- on connect: GET ${ARK_ONECRAFT_BASE_URL}/api/jst/score?externalUserId=<id>
  (Bearer ARK_ONECRAFT_API_KEY); 502 ARK_UNREACHABLE | ARK_REJECTED |
  ARK_INVALID_SCORE on failure paths
- validates jobs/skills/talent as integers 1..10, re-scores SERVER-SIDE
  (scoreJst), inserts source='ark_onecraft', returns JstSummary
GET /me/jst      getMyJst      // JstAssessment.source now a required enum field
POST /me/jst     submitMyJst   // self-assessment → source='self_assessment'
GET /me/ascension getMyAscension
POST /me/onboarding/reader-code claimReaderCode
```

Front-end (current change)

`artifacts/command-centre/src/pages/ascension.tsx` adds a primary **"UPLOAD JST SCORE FROM ARK.ONECRAFT"** button above the self-assessment (reframed "OR SELF-ASSESS (INTERIM)"). Uses the generated `useImportJstFromArk` hook. Not-connected detection is robust — checks `ApiError.status === 503` and the body's `data.code === "ARK_NOT_CONFIGURED"` (the generated client surfaces the response body on `err.data`), and shows a non-destructive "ARK.ONECRAFT — coming soon" toast. Imported scores show a "via ARK.ONECRAFT" source badge.

Deployment topology

- Shared reverse proxy routes by path (`/api` → api-server, `/` → command-centre). **Never call service ports directly — always `localhost:80`.**
- Published over HTTPS on `$REPLIT_DOMAINS`. Scheduled Deployments drive two cron endpoints (`send-weekly-digest`, `reset-harness-limits`) via the `scripts` package.

Observability & cost control

- pino structured logs (`req.log` in handlers, singleton `logger` elsewhere).
- Optional Sentry (`SENTRY_DSN` / `VITE_SENTRY_DSN`, no-op when unset).
- Monthly LLM cost cap enforced per engine route (`402 COST_CAP_EXCEEDED`); live SUM over `harness_engine_runs.cost_usd`, tier defaults + per-subscriber override.

Testing strategy

- Offline cross-provider fixture replay (30 tests, ~15s); cache keys normalise UUIDs / ISO timestamps. `test:live` (`RECORD=1`) refreshes fixtures (~10 min) and runs nightly. JST scoring covered by `test/ascension-jst.test.ts`.

Threat-model summary

- **AuthZ:** all reads/writes scoped to the authenticated user; org routes validate supplied `userIds` against org membership (anti cross-org fuzzing).
- **SSRF:** external fetches are env-driven (ARK host, Sphinx host); badge URL verification is SSRF-hardened. Hardening backlog: enforce `https://` + host allowlist on the ARK base URL.
- **Webhook integrity:** Stripe signature + idempotency PK; handler throw rolls back the idempotency row so Stripe can legitimately retry.
- **Secrets:** integration API keys stored AES-256-GCM (script-derived from `SESSION_SECRET`); no plaintext at rest.
- **Drift:** PFP `counts` / `verdict` recomputed server-side; F8 `DRIFT_GATE`.

Env (delta for this change)

`ARK_ONECRAFT_BASE_URL` + `ARK_ONECRAFT_API_KEY` (both required to activate the JST import; absence → typed 503). Documented in `replit.md` → Env → Integrations & ops.

Appendix A — ATLAS-J Crystallised View (typed)

The `atlas-pdd-v1` JSON projection: per-phase prompts `P-<PHASE>-NNN`, CLASS A (foundational) / B (feature) / C (enhancement).

```
{
  "schemaVersion": "atlas-pdd-v1",
  "title": "ATANDA Command Centre — full system + ARK.ONECRAFT JST import",
  "summary": "Authenticated portal over the FORGE.BONSAI HARNESS (F1-F8 + ATLAS J + PFP + DE-SPC), tiered Stripe billing, org seats, onboarding",
  "prompts": [
    { "id": "P-RED-001", "phase": "RED", "title": "DB + Drizzle schema", "operation": "Provision Postgres, one-file-per-table schema, barrel e"},
    { "id": "P-RED-002", "phase": "RED", "title": "Clerk auth bridge", "operation": "Mount Clerk proxy before body parsers; JIT local user + su"},
    { "id": "P-RED-003", "phase": "RED", "title": "Stripe + webhook idempotency", "operation": "Raw webhook route before JSON parser; PK idemp"},
    { "id": "P-ORANGE-001", "phase": "ORANGE", "title": "Portal shell + pricing", "operation": "Landing, /pricing, tiered checkout, session sho"},
    { "id": "P-YELLOW-001", "phase": "YELLOW", "title": "HARNESS F1-F7 pipeline", "operation": "Ordered atomic-prompt engines → certified MVP-1"},
    { "id": "P-YELLOW-002", "phase": "YELLOW", "title": "F8 Code DJ + PFP drift gate", "operation": "Scaffold CODEBASE_BUNDLE; enforce DRIFT_G"},
    { "id": "P-YELLOW-003", "phase": "YELLOW", "title": "Ascension Protocol + JST", "operation": "Book-anchored 13-rung ladder seeded by JST se"},
    { "id": "P-YELLOW-004", "phase": "YELLOW", "title": "ARK.ONECRAFT JST import", "operation": "Env-gated import endpoint + upload CTA; source"},
    { "id": "P-GREEN-001", "phase": "GREEN", "title": "Public verify + exemplars", "operation": "SPARTAN verification URL, exemplar fork-to-se"},
    { "id": "P-BLUE-001", "phase": "BLUE", "title": "Cost cap + audit + offline tests", "operation": "requireCostBudget, activity log, 30-est"},
    { "id": "P-INDIGO-001", "phase": "INDIGO", "title": "Orgs + per-seat tiers + notifications", "operation": "Org tenancy, effectiveTier, weel"},
    { "id": "P-VIOLET-001", "phase": "VIOLET", "title": "Sphinx + ARK federation (deferred)", "operation": "Two-way marketplace + identity fed"},
  ],
  "stack": ["pnpm-monorepo", "Node.js 24", "TypeScript 5.9", "Express 5", "Clerk", "Stripe", "PostgreSQL", "Drizzle ORM", "Zod v4", "Orval", "P"},
  "routes": ["/", "/pricing", "/ascension", "/cartridge", "/quests", "/me/costs", "/api/harness/f1..f8", "/api/harness/atlas-crystallise", "/ap"},
  "deployTarget": "express-replit (api-server) + react-vite-static (command-centre) behind the shared path-routing proxy, published over $REPL",
}
```